

Computing for Analytical Work

Session 2 — What is running?

Block 3 — Python interpreters, packages, imports, and environments

Knowledge check 2 — 10 minutes, closed book, 3 questions

- Covers Session 1 and Homework 1: paths, working directory, terminal, `git status`
- Laptops closed. Paper only.
- Three short questions. Diagnostic, not tricky.
- Worth 5% of the final grade
- **This block's lecture is 35 minutes**, not 45 — the check comes out of lecture, never out of lab
- Hand it in when the timer goes. Then we begin.

This block in one sentence

"Python" means a particular interpreter with a particular set of installed packages.

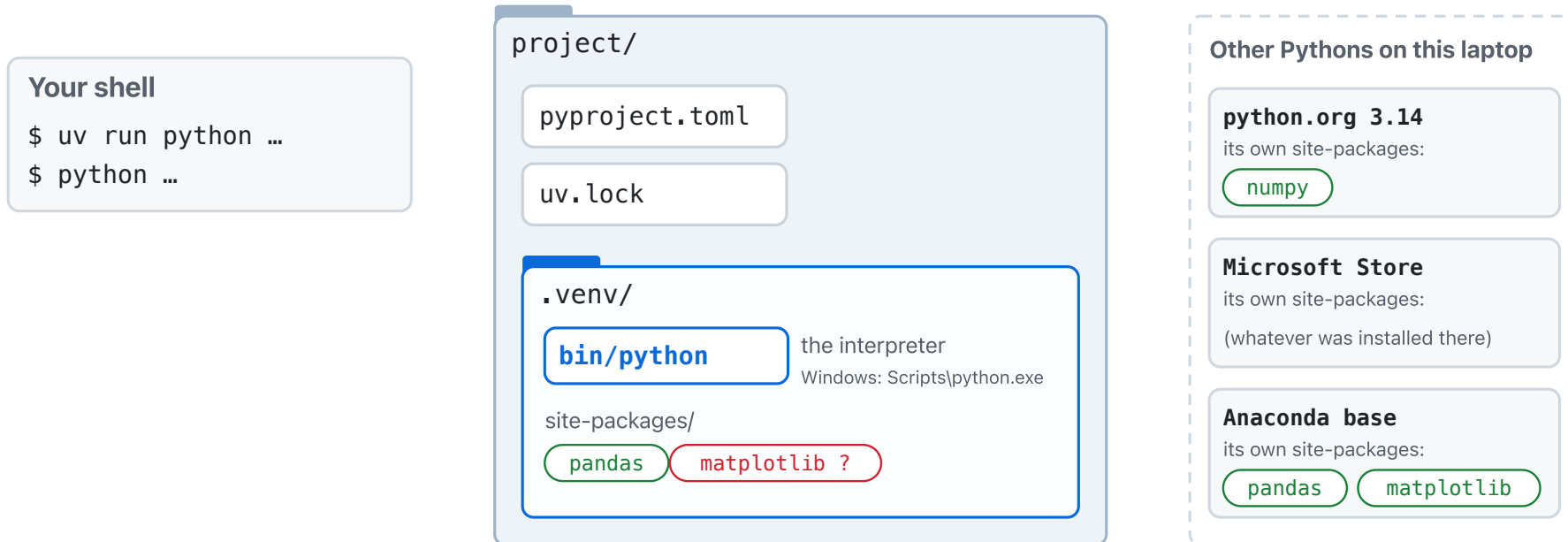
Not a language. Not an icon. A specific program, on a specific path, with a specific list of packages it can see.

What we cover

- The reveal: what `uv run` has been doing all week
- Interpreter, script, notebook
- Packages, `import`, and `sys.path`
- Installing into a specific environment; why installs "fail"
- Many Pythons on one machine, and how to check which is active
- `uv` : the one workflow this program uses
- Git thread: `git add`, `git commit`
- AI thread, stretch, and Lab 3

You have been typing `uv run` for a week

`uv run python scripts/summarize.py` — in the setup check, Lab 1, Lab 2, Homework 1. Every README said: *"You will learn what `uv run` does in Session 2. For now, type it exactly."* It is Session 2. This map is the answer.



Get the block's project: clone Lab 3 now

In your terminal (Git Bash on Windows, Terminal on macOS), in the folder where you keep course work:

```
git clone https://github.com/earino/ecbs5293-lab03-environments.git  
cd ecbs5293-lab03-environments
```

- Every command you *type* in this block is typed **there**: that terminal, that folder. Slides you only watch say so.
- It is the Lab 3 starter. Leave its README and its script alone until the lab; the lecture only looks at the project's environment.

Interpreter, script, notebook

- **Interpreter** — a program, one executable file, that reads Python code and runs it
- **Script** — a `.py` file the interpreter reads top to bottom, once, then exits
- **Notebook** — a long-running interpreter (the *kernel*) you feed one cell at a time (Block 4, this afternoon)

In that terminal, in the Lab 3 folder:

```
uv run python -c "import sys; print(sys.executable)"
```

That prints the path of the interpreter that ran the line. It is a file. You can `ls` it.

What a package is

- A folder of Python code somebody else wrote, plus metadata
- `pandas`, `matplotlib`, `jupyter` — each is a folder on disk, installed *somewhere*
- Installed means: copied into a folder **that one interpreter knows to look in**

Same terminal, same folder:

```
uv run python -c "import pandas; print(pandas.__file__)"
```

That prints where pandas lives. Note the path. It will matter in five slides.

What `import` actually does

```
import pandas
```

1. The interpreter takes a list of folders called `sys.path`
2. It walks that list, in order, looking for a folder or file named `pandas`
3. First match wins. No match: `ModuleNotFoundError`

`import` is a **search**. It is not a download, and it is not a question about your laptop.

`sys.path` belongs to *that* interpreter

`import pandas` is a search through one interpreter's `sys.path`, in order. First match wins.

sys.path of `.venv/bin/python`

on Windows: `.venv\Scripts\python.exe`, `.venv\Lib\site-packages\`

- 1 `project/`
- 2 `.venv/lib/python3.13/site-packages` `pandas ✓`

found: stop, import it

sys.path of `/usr/bin/python3`

on Windows this would be e.g. `C:\Python314\python.exe`

- 1 `/usr/lib/python3.9/`
- 2 `/usr/lib/python3.9/site-packages/`
- 3 `/usr/lib/python3/dist-packages/` `no pandas`

end of the list: `ModuleNotFoundError`

Same code. Two interpreters. Each searched its own list and told the truth.

So "is pandas installed?" is an incomplete question. **Installed for which interpreter?**

Installing into a specific environment

- An **environment** is one interpreter plus the folders it searches. That is all it is.
- Installing a package puts it into **one** environment
- This course: every project has its own environment, in `.venv/`, inside the project folder

```
uv add <package>
```

Every *external* package your code imports needs a line in `pyproject.toml`. `uv add` installs it *and* writes the line. Python's own built-in library — `sys`, `pathlib` — is already there; it needs nothing. Watch; you type `uv add` in the lab.

Why installs "fail"

1. This Python installed it

```
(other) anna@laptop project $ pip install matplotlib
```

```
- - - pip's download and cache lines, elided - - -
```

```
Installing collected packages: six, pyparsing, pillow, packaging, numpy, kiwisolver, fonttools, cyclor, python-dateutil, contourpy, matplotlib
```

```
Successfully installed contourpy-1.4.0 cyclor-0.12.1 fonttools-4.65.0 kiwisolver-1.5.1 matplotlib-3.11.2 numpy-2.5.3 packaging-26.3 pillow-12.3.0 pyparsing-3.3.2 python-dateutil-2.9.0.post0 six-1.17.0
```

```
(other) anna@laptop project $ █
```

2. This Python looked for it — one minute later, same terminal

```
(other) anna@laptop project $ uv run python -c "import matplotlib"
```

```
warning: `VIRTUAL_ENV=/Users/Shared/anna/other` does not match the project environment path `.venv` and will be ignored; use `--active` to target the active environment instead
```

```
Traceback (most recent call last):
```

```
File "<string>", line 1, in <module>
```

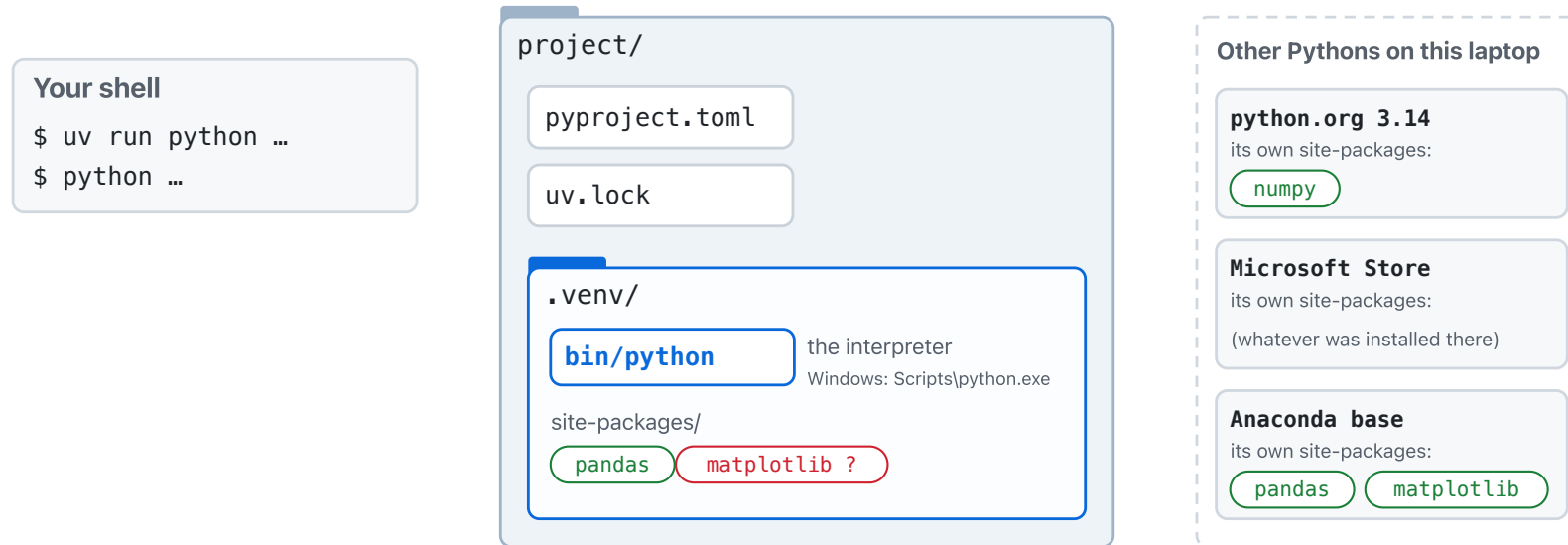
```
import matplotlib
```

```
ModuleNotFoundError: No module named 'matplotlib'
```

```
(other) anna@laptop project $
```

`pip` is a package installer. The `pip` you type belongs to one particular Python, and installs into that one — that is the whole problem. The install did not fail; it **succeeded, somewhere else**.

Many Pythons on one machine



Each has its own `sys.path`. Nothing coordinates them. **Which one runs is decided by which one you invoke.**

Bare `python` is resolved by the shell walking a list of folders called `PATH` — first match wins, and every installer puts its own folder at the front. **In Git Bash, always pass an argument:** bare `python` either hangs with no error or drops you into Python's `>>>` prompt.

Predict: two Pythons, side by side

```
python -c "import sys; print(sys.executable)"  
uv run python -c "import sys; print(sys.executable)"
```

My laptop. A terminal *outside* VS Code, standing in a project folder like yours, thirty seconds apart. **Will these two commands print the same path?**

Write down *yes* or *no*, and one reason. Do not type yet.

(The recording then asks each of those two Pythons the same `import matplotlib .`)

Two Pythons — what happened

```
anna@laptop project $ python -c "import sys; print(sys.executable)"
/opt/homebrew/Caskroom/miniconda/base/bin/python
anna@laptop project $ uv run python -c "import sys; print(sys.executable)"
/Users/Shared/anna/project/.venv/bin/python3
anna@laptop project $ python -c "import matplotlib; print(matplotlib.__file__)"
/opt/homebrew/Caskroom/miniconda/base/lib/python3.12/site-packages/matplotlib/__init__.py
anna@laptop project $ uv run python -c "import matplotlib; print(matplotlib.__file__)"
Traceback (most recent call last):
  File "<string>", line 1, in <module>
    import matplotlib; print(matplotlib.__file__)
    ^^^^^^^^^^^^^^^^^
ModuleNotFoundError: No module named 'matplotlib'
anna@laptop project $
```

Two paths. The first is whatever *my* shell found; the second is *this project's* interpreter. Then the same import, asked of each.

What you just saw, on the map



The project's own environment, versus every other Python on the laptop. That contrast is the whole block.

Part 2 — **uv**: the one workflow

The three commands

```
uv sync          # install everything the project declares
uv add pandas    # declare and install a new dependency
uv run python script.py  # run a script inside the project's environment
```

Typed in your terminal, always from inside the project folder. That is the whole workflow for this program. **Watch this one; nothing to type** — `script.py` is a placeholder, not a file in this repo.

pyproject.toml and uv.lock, in plain words

```
pyproject.toml

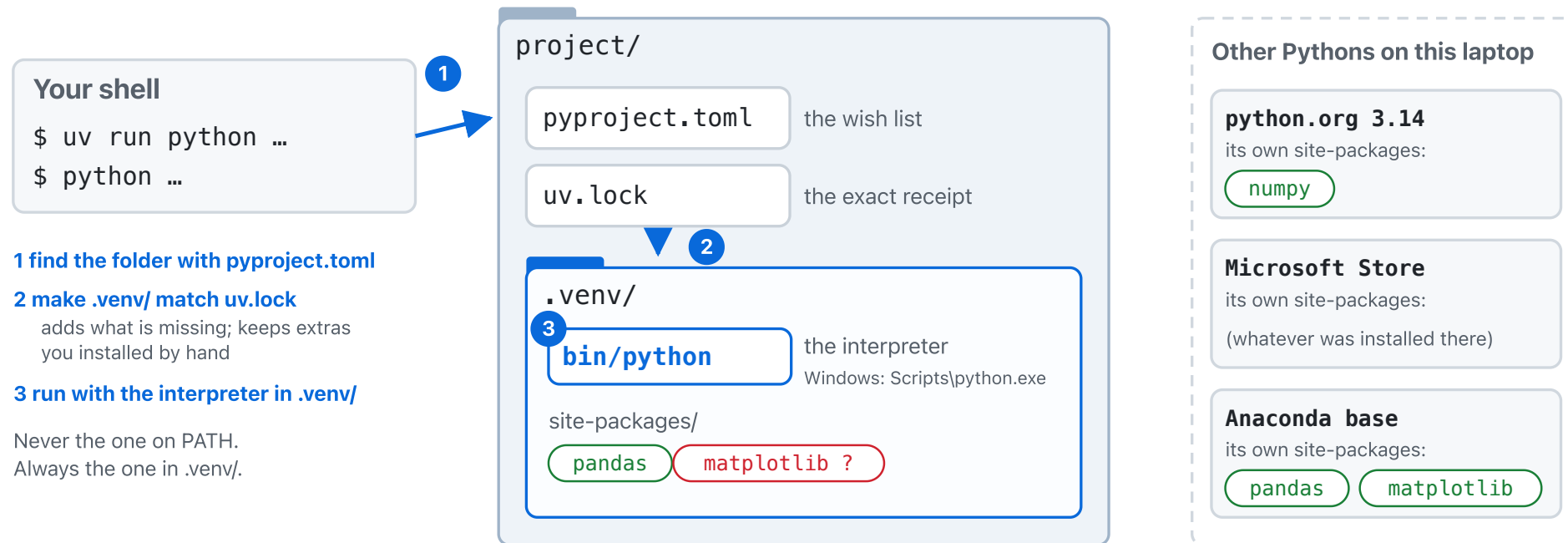
[project]
name = "ecbs5293-lab03-environments"
version = "0.1.0"
description = "ECBS5293 lab 3: it imports on ..."
requires-python = ">=3.10"
1 dependencies = [
    "pandas",
]
```

```
uv.lock (one of its entries)

[[package]]
2 name = "pandas"
  version = "3.0.5"
  source = { registry = "https://pypi.org/simple" }
3 dependencies = [
  sdist = { ..., hash = "sha256:dca3734d6ab7c906e6730f..." }
```

- 1 The wish list: what this project needs, by name. uv add writes the line for you; you do not hand-edit it.
- 2 The receipt: the exact version that satisfied the wish, for every package, including ones you never named.
- 3 A hash of the bytes: uv sync installs the same thing on every machine, and refuses anything else.

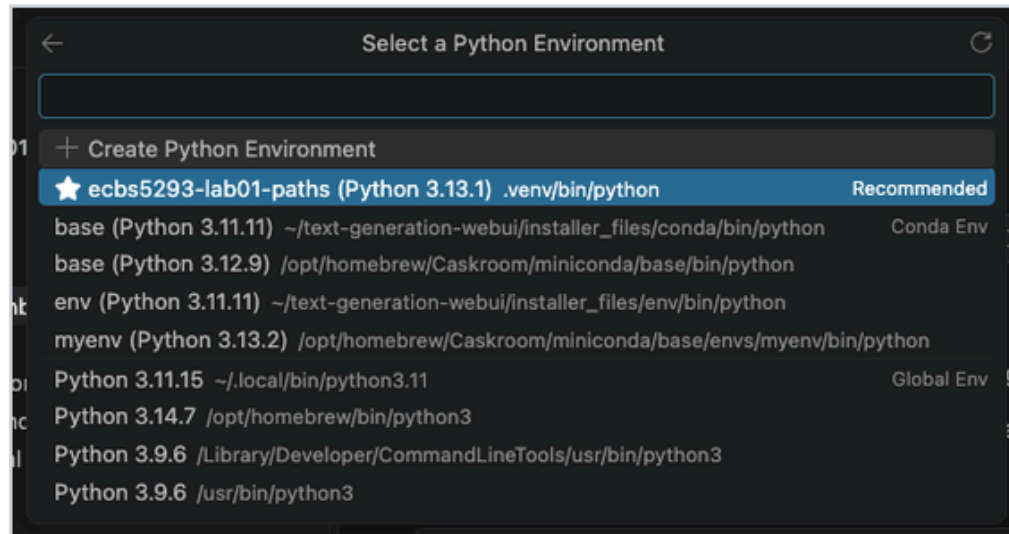
So what was `uv run` doing?



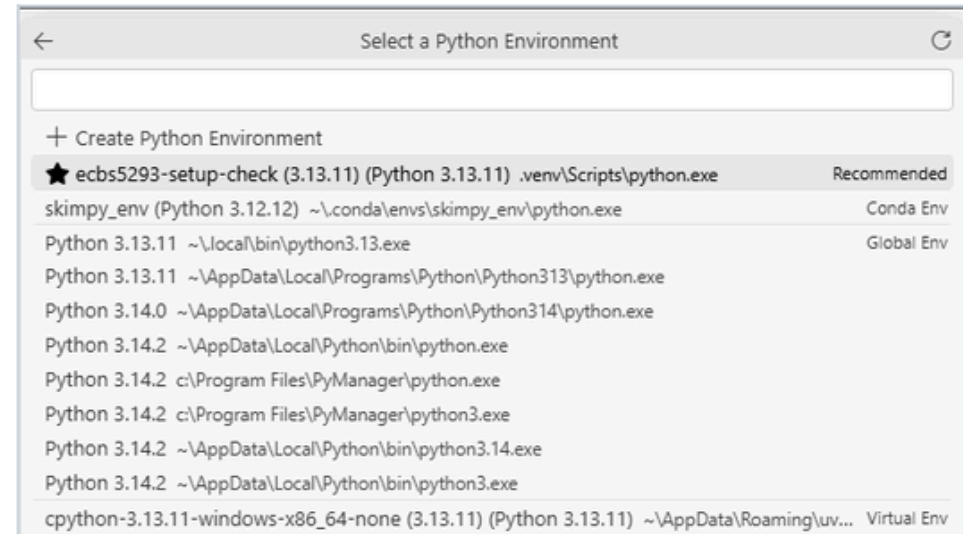
`uv run` 's quiet sync **adds** what is missing and leaves hand-installed extras alone. A plain `uv sync` makes `.venv` match the declaration **exactly** — and removes them.

The notebook standard: the project's own kernel

macOS: `.venv/bin/python`



Windows: `.venv\Scripts\python.exe`



Above: VS Code's kernel picker, macOS on the left, Windows on the right.

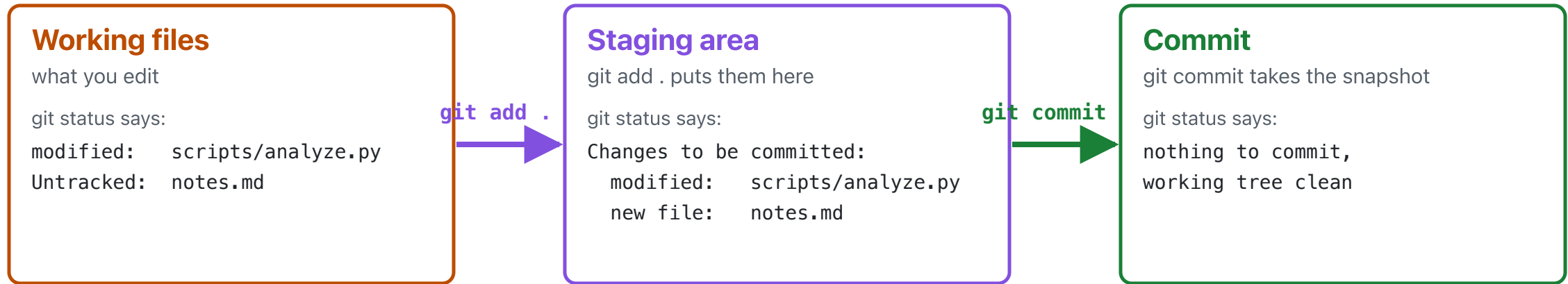
1. Open the **project folder** (never the lone `.ipynb`).
2. Kernel picker → the interpreter inside `.venv/`.
3. Verify: `import sys; sys.executable` in a cell prints a path inside *this project's* `.venv/`.

In the wild: `requirements.txt`

- Older repos, blog posts, and other courses ship `requirements.txt` instead of `pyproject.toml`
- Same job, older file: *make this project's packages available to its interpreter*
- You will not write one in this course. You need to be able to **recognise** one.

If a README says `pip install -r requirements.txt` and there is no such file, that README is wrong.

A whiff of Git — save a checkpoint



git diff compares only these two. Staged work and new files are invisible to it.

```
git status          # what changed?
git add .           # stage it: "include these in the next snapshot"
git commit -m "Fix the output path in analyze.py; add notes"
```

A commit is a **checkpoint after a working fix**. Commit when it works. Not before, not "at the end". **Watch now; you type these in the lab.**

After you commit: `git status` first, then `git diff`

```
anna@laptop project $ git status --short
M scripts/analyze.py
?? notes.md
anna@laptop project $ git add .
anna@laptop project $ git diff
anna@laptop project $ git status --short
A notes.md
M scripts/analyze.py
anna@laptop project $ git commit -q -m "Fix the output path in analyze.py; add notes"
anna@laptop project $ git status
On branch main
nothing to commit, working tree clean
anna@laptop project $
```

- `git status` covers everything: unstaged edits, staged work, and files Git has never seen. "nothing to commit, working tree clean" is the completeness check.
- `git diff` alone shows **unstaged edits to tracked files**. Staged work and new files are invisible to it.

AI thread — ask for evidence, not for a fix

Good questions for an assistant today:

- *"What evidence distinguishes 'package not installed' from 'installed in a different environment'?"*
- *"What does `uv sync` actually do? What is `pyproject.toml` ? How is `uv` different from `pip` or `conda` ?"*
- *"Here is `sys.executable` from my terminal and from my notebook. They differ. What does that mean?"*

Then run the inspection yourself and paste the real output.

The active interpreter decides what is true

- AI can suggest causes. It cannot see your machine.
- `sys.executable` shows which interpreter answered; printing `sys.path` shows where it searched.
- If the assistant says "it's installed" and the interpreter says `ModuleNotFoundError`, the interpreter is right.

Sources of truth today: the filesystem, the interpreter, `uv`'s output. Everything else is a hypothesis.

5-minute stand-and-stretch

Stand up. Leave the laptop. Back in five.

Lab 3 — It imports on my machine, but not here

First 15 minutes: no AI. Chat windows and assistant panels closed; VS Code itself stays open. Terminal, filesystem, and `uv` output only. Search engines are fine.

Why: every failure today is one an assistant diagnoses instantly. The reflex we want is *"check which interpreter,"* not *"ask what to check."* The exam has no chat window. Solve it in five minutes and write the note; the fifteen minutes are for looking, not for suffering.

TAs will ask you to close a window. That is not a penalty.

The symptom, and how to run it

```
anna@laptop ecbs5293-lab03-environments $ uv run python scripts/plot.py
Traceback (most recent call last):
  File "/Users/Shared/anna/ecbs5293-lab03-environments/scripts/plot.py", line 8, in <module>
    import matplotlib
ModuleNotFoundError: No module named 'matplotlib'
anna@laptop ecbs5293-lab03-environments $
```

In your terminal, in the Lab 3 folder you cloned at the start of the block:

```
uv sync
uv run python scripts/plot.py
```

The README says `pip install -r requirements.txt`. There is no such file. Your job: which interpreter is looking, what the project *declares*, and why they disagree. Edit files in VS Code (open the folder; trust it if asked). At home: the README begins with the `git clone` line.

What you produce, and the last ten minutes

1. `pyproject.toml` declares the missing dependency (`uv add` , not a hand edit) — **checked** by a plain `uv sync` , then `uv run python scripts/plot.py` runs clean and writes the PNG. Sync first, then run.
2. README setup instructions are correct for *this* repo
3. `DIAGNOSIS.md` : symptom, cause, evidence (paste real output), change, verification
4. One commit of the working state; `git status` reports a clean tree afterward
5. At minute 33: explain it to a neighbour, about a minute, no notes; the what-if about environments is on the next slide. Checkpoint on Moodle before you leave.

Reset instructions are in the README; both resets destroy work.

The last ten minutes — explain it to a neighbour

Speaker, about a minute, pointing at the screen, not reading the note: symptom → cause → evidence → change → verification.

Listener asks:

1. Show me the evidence.
2. Why did it fail — not just where?
3. Suppose you had fixed this with `uv pip install matplotlib` instead of `uv add`, and committed. A classmate clones your repo and runs `uv sync && uv run python scripts/plot.py`. What happens, and why?

Swap. Unsure, or you disagree? **Hand up.** Then the answer, for everyone. Before you leave: `DIAGNOSIS.md` to the Lab 3 checkpoint on Moodle.